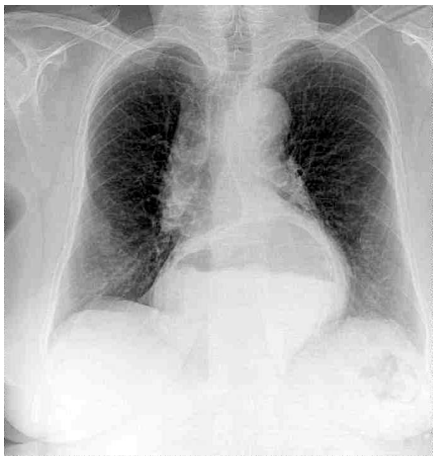
X-ray



An X-ray photograph of lungs.

X-ray is part of the electromagnetic spectrum, with wavelength shorter and energy higher than ultraviolet radiation. The wavelength of X-ray covers a wide range. Radiation with shorter wavelength (higher energy) than X-ray is called Gamma ray.

X-rays can go through many solid materials. They can be produced by hitting metal with fast-moving electrons. X-ray photography is used in medicine to see bones and cancer, and other things inside bodies. A Computed tomography scanner combines an X-ray machine and computer to construct a three dimensional (3D) picture.

X-rays are photons, tiny packets of energy that interact with atoms in the body. The way they interact depends on the wavelength (energy level) of the x-rays. X-rays with low-level energies ("soft" x-rays) go through the Photoelectric effect, mid-level energies go through Compton scattering, and high-level energies ("hard" X-rays) go through pair production. Medical x-rays can experience the Photoelectric effect and sometimes Compton scattering. Radiation therapy that treats cancer use Compton scattering and sometimes Pair production.

There are small amounts of X-rays in the atmosphere. Like other ionizing radiation, X-rays can change living cells. This means that exposing the human body to high doses of x-rays for a long time should be avoided. Changing living cells can be dangerous. It can cause cancer and other diseases. It can also be good, as cancer cells can be killed that way. Like with any treatment, there are side-effects.

X-rays cannot see soft tissue very well. To detect tumors, other imaging technologies are used; such as Magnetic resonance imaging.

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